Channel-Aware Training, Not Closed-Loop Control, Drives Robust Regeneration in Neural Cellular Automata

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Abstract

We set out to test whether closed-loop control of global chemical signals makes regenerative Neural Cellular Automata (NCAs) more robust, and built the evaluation to prove it: recurring multi-block lesions larger than the perception radius, a small evolved controller for three broadcast modulator channels, and held-out damage seeds. On a single trained model, the experiment looked like a success. However, crossing five independently trained parent seeds (two independent controller evolutions each; preregistered) dissolves that success into three findings. First, the robustness we had attributed to control is mostly a property of training: parents trained with channels present beat unmodulated siblings in five of five seeds (median final-Hamming reduction 0.008, up to ≈ 0.14 on the most fragile parent) with modulation pinned to neutral. Second, evolved controllers add little on top: on four parents the effect is absent or noise-level; on the fragile parent both evolutions reproducibly help (0.035–$0.038 \rightarrow 0.029$–0.030, survival $0.975 \rightarrow 1.00$); yet its tonic vector is nearly identical (cosine 0.99) to a sibling's that does not help, so the benefit is an interaction with parent dynamics, not a distinctive operating point. All ten controllers emit flat, nonzero, parent-specific constants with no lesion-locked response. Third, these constants are organism-locked: a single-donor probe penalizes five of five siblings (lethally in two), a full 5×5 transfer matrix shows most transfers harmful and none better than the recipient's own controller, and injecting each donor's tonic constant directly, no controller at all, reproduces its transfer outcomes in 46 of 50 cells, including all lethal ones. Single-parent evaluation would have approved all of it, making the attribution protocol itself the transferable artifact.

1 Introduction

Growing Neural Cellular Automata (GNCAs) grow a target morphology from a single seed and regenerate it after damage using one local, differentiable update rule [1]. The catch is perception: a cell sees only its immediate neighborhood, so wounds larger than a few cells contain tissue with no signal to integrate, and a severed fragment has no way to know what it was part of. The natural remedy is a global signal. Prior work, our own single-parent study included (Appendix H), adds global modulator channels and evolves release policies for them, reporting the outcome on one trained model.

That single-model habit is the problem this paper is about. When a modulated NCA beats an unmodulated one, the channels being *present during training*, the controller *acting at run time*, and whatever *transfers* to another organism are confounded. Our single-parent study learned this the hard way: it credited closed-loop control for a $2\times$ improvement that vanished under cross-parent

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testing, where the controller transferred lethally to siblings while zero-modulation channel parents beat the unmodulated baseline in every seed (Appendix C). The improvement was real, but the attribution was wrong.

We therefore ran the preregistered attribution study that the original design needed with primary statistic and decision thresholds fixed before data collection. Five parent seeds, each trained from scratch as a K=0 pair and a channel-aware K=3 pair; *two* independent controller evolutions per channel parent; four conditions per seed, all on held-out damage seeds; then, post-hoc, a full 5×5 transfer matrix and a tonic-transplant condition (Section 3; preregistrations and analysis scripts timestamped before the runs).

Contributions.

1. A preregistered robustness-attribution protocol for regenerative NCAs: adversarial damage calibrated to exceed perception, crossed with parent-seed variation, decomposed into E_{train}, E_{ctrl}, E_{transfer}, with calibration probes that keep the evolution landscape interpretable (Appendix F).
2. The attribution, settled causally: training with channels is the robust cause (5/5 seeds); the controller effect is parent- dependent (absent or noise-level on four parents, reproducible on the fragile one); the evolved artifact is a parent-specific tonic constant that penalizes every sibling it meets (lethally in two), and injecting the constant alone reproduces its transfer outcomes: the matrix (23/40 harmful, none beating the recipient's own) closes the case.

2 Related Work

Neural cellular automata. GNCAs grow patterns from a seed and regenerate after damage [1], with extensions to self-classification [2] and textures [3]. All rely on local perception, and regeneration degrades beyond the perception radius; the regime that our benchmark exploits. Signal channels [4], goal conditioning [5], and information-dynamical analyses of self-maintenance [6] all treat the signal as fixed input or emergent byproduct instead of a controlled output; we ask what is actually helping when such a channel *appears* to help.

Search over evaluation regimes. Our design follows a concern shared by novelty search [7] and quality-diversity optimization [8]: a fixed objective can misdirect search and hide failure modes. Environment-generation methods co-evolve challenges with solutions. POET [9], PAIRED [10], ACCEL [11]; Our benchmark does not yet evolve the damage distribution and serves as the stationary baseline such co-evolution can be judged against.

3 Experimental design

Damage regime. Recurring multi-block lesions; every 150 steps, n=4 contiguous 16×16 blocks at seeded positions, T=2000 (the information-isolation regime of Appendix E). Damage seeds 0–7 drive evolution; held-out seeds 10000–10007 drive all reported numbers; the schedule is deliberately adversarial (Appendix F).

Parents and controllers. Per seed s we train a K=0 and a channel-aware K=3 parent from scratch (Appendix D); the K=3 parent's three global modulator channels are the only non-local pathway, each carrying a *tonic* (slow baseline) and *phasic* (event-triggered, fast-decaying) component: the distinction from biological neuromodulation. A 259-parameter controller (4→32→3, $\tanh$) reads four target-free grid statistics and sets release every 10 steps; each parent gets *two* independently evolved controllers via CMA-ES [12] in Evosax [13] (population 64, σ_0=0.01, 300 generations, different evolution seeds, event-weighted Hamming objective; Appendix G).

Preregistered comparison. All conditions run on the same held-out damage seeds (5 condition seeds × 8 damage seeds) from a shared t=0 state grown by the K=3 parent. The primary statistic is $\Delta_s = H(K{=}3, m{=}0, s) - H(K{=}3, \text{own ctrl}, s)$, positive when the evolved controller helps *its own* parent. Fixed before data collection: *controller effect supported* if $\Delta_s > 0$ in $\geq 4/5$ seeds with sign-consistent differences; *channel-training effect supported* if $H(K0, s) - H(m0, s) > 0$ in

$\geq 4/5$; *transfer failure supported* if the July controller underperforms the own-controller substantially in $\geq 4/5$. Effects are reported per seed (median and range); no significance claims at five seeds. The defense study reported here fixed the two-evolutions-per-parent rule in its preregistration before launch (`experiment_results/20260818_evoseed_defense/PREREGISTRATION.md` in the repository history). Controller-output (m_t) series distinguish tonic calibration from event-locked policy (Appendix B). A post-hoc follow-up evaluates the full $5{\times}5$ transfer matrix among the defense parents (both controller replicas, 8 damage $\times$ 3 condition seeds per cell; Table A3) plus a *tonic transplant* injecting each donor's realized mean m_t as a constant.

4 Results

Table 1 reports the preregistered decomposition; the full per-seed, per-condition numbers behind it are in Appendix A.

Table 1: Final Hamming (mean over 5 condition seeds $\times$ 8 held-out damage seeds). Survival in parentheses where it departs from 1.00. Training helps in 5/5 seeds; the controller effect is parent-dependent.

Parent seed	$K{=}0$	$K{=}3$, $m{=}0$	Own ctrl (2 runs)	Interpretation
0	0.035–0.036	0.029	0.029–0.030	channel effect; no control effect
1	0.163–0.175 (surv. 0.05–0.08)	0.035–0.038 (surv. 0.975)	0.029–0.030 (surv. 1.00)	channel effect + replicated tonic benefit
2	0.030	0.018–0.021	0.022–0.026	channel effect; controller slightly worse
3	0.049–0.052	0.028–0.029	0.032–0.033	channel effect; controller slightly worse
4	0.042–0.045	0.033	0.034–0.035	channel effect; no control effect

Outcome. *Channel-training supported* (positive in 5/5 seeds; bar $\geq$4/5); *transfer failure supported* (penalty 5/5, lethal in 2). Under the preregistered rule the controller effect is classified as *parent-dependent*, preregistered **Outcome C**: noise-level on four parents, reproducibly beneficial on the fragile one (both evolutions: 0.035–0.038 $\rightarrow$ 0.029–0.030, survival 0.975 $\rightarrow$ 1.00).

What the controller actually emits. All ten evolved controllers emit a *constant at a nonzero level*: within-rollout per-channel std of m_t is 0.0002–0.0046, tonic with no lesion-locked response in 10/10 runs, while channel means sit at parent-distinct offsets (Figure A1). Evolution neither collapses to neutral output nor discovers a policy: it finds a parent-specific tonic calibration (whose predictive limits are discussed in Section 5).

Cross-parent transfer matrix. The full $5{\times}5$ matrix among the defense parents, both replicas, three condition seeds per cell, 40 off-diagonal cells (Table A3, Figure 1), 23/40 transfers are harmful (survival < 0.9 or final Hamming worse than the recipient's zero-output baseline by > 0.005), 15 are indistinguishable from zero-output, and 2 beat it. No foreign controller beats the recipient's own controller beyond noise. Lethal transfer replicates across controller replicas and condition seeds: donor s3 kills recipient s2 in both replicas (survival 0.00); donor s0 on recipient s1 is lethal in e1 and near-lethal in e2 (Figure 1A).

Tonic transplant. Injecting each donor's realized mean m_t as a constant in place of its controller (both replicas, three condition seeds) reproduces the full controller's outcome in 36/40 off-diagonal cells, including every lethal transfer (the four mismatches are magnitude differences inside already-lethal cells), and matches the recipient's own controller in all 10 diagonal cases (46/50 overall; Appendix A). The transferable component is thus the tonic setpoint; the controllers' small dynamic residual did not measurably contribute.

5 Discussion

Which component causes robustness. Channel-aware training, not run-time control. Parents that grew up with global channels beat their unmodulated siblings on every seed; the channels are a

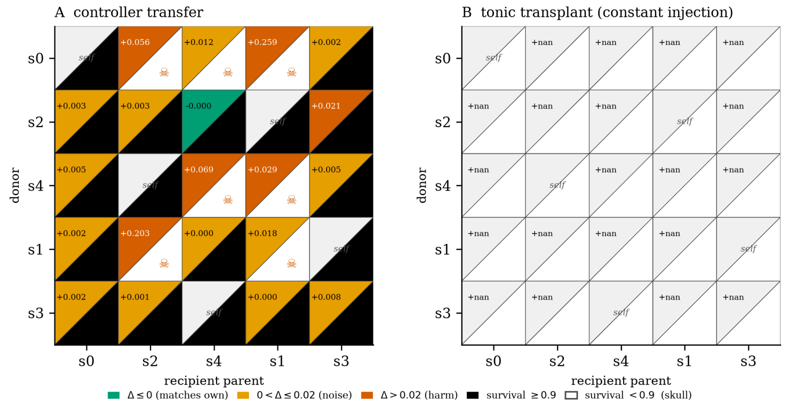

Figure 1: Transfer redesigned for attribution (replica-averaged; per-cell Δ from the recipient's own controller; survival as the split lower triangle; rows/columns cosine-ordered so the aligned pair s4–s1 borders the diagonal and the anti-aligned pair s3–s2 sits at opposite ends). Panel A: controllers. Panel B: the same matrix with each donor's tonic constant injected instead, matching A in 46/50 cells, every lethal one included. Values: Table A3.

developmental scaffold: more robust even with no run-time signal. What evolution adds is narrow: nothing measurable on four parents; on the fragile parent, a final stabilization that both independent evolutions find. Since that parent's tonic vector is nearly identical to a sibling's that gains nothing from it, the benefit is not a property of the tonic.

Why cross-parent transfer fails, and when it does not. The full matrix (Table A3) sharpens the single-donor probe: no foreign controller beats the recipient's own controller, most transfers are harmful (23/40), and the only beneficial ones stay within the tonic-aligned pair (s4→s1). The tonic transplant makes it causal: what transfers, for good or ill, *is* the constant (36/40 off-diagonal, 10/10 diagonal, every lethal cell included). Each controller's output is a calibration against its own parent's channel weights, so the same release level drives different dynamics in a sibling; evolution found an operating point for one organism, not a modulation law.

Limitations and future work. Five parents, two evolutions each; one morphology; one damage family. Only one parent in five proved controller-responsive. Next: multi-parent evolution and non-stationary, diversity-driven damage co-evolution, for which this benchmark is the controlled baseline.

Conclusion. In regenerative NCAs under adversarial recurring damage, robustness comes from channel-aware training (5/5 seeds); evolved modulation is a parent-dependent tonic calibration, noise-level on most parents, beneficial on the fragile one, parent-locked under transfer. Attribute robustness across model seeds, not only damage seeds.

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A Full per-seed, per-condition results

Table A1: Final Hamming (mean ± SD, 5 condition seeds × 8 held-out damage seeds) and survival for each of the ten runs (5 parent seeds × 2 independent controller evolutions) under hard recurring multi-block damage, T=2000. Conditions: K=3 parent with modulation pinned to zero; the parent's own evolved controller; the K=0 parent.

	K=3, m=0		K=3, own ctrl		K=0	
Run	H	surv	H	surv	H	surv
s0_e1	0.0290±.0135	1.000	0.0303±.0134	1.000	0.0361±.0137	1.000
s0_e2	0.0294±.0144	1.000	0.0292±.0136	1.000	0.0354±.0109	1.000
s1_e1	0.0384±.0266	0.975	0.0286±.0177	1.000	0.1628±.0389	0.075
s1_e2	0.0353±.0256	0.975	0.0301±.0210	1.000	0.1746±.0356	0.050
s2_e1	0.0183±.0145	1.000	0.0225±.0149	1.000	0.0299±.0178	1.000
s2_e2	0.0208±.0170	1.000	0.0258±.0161	1.000	0.0296±.0174	1.000
s3_e1	0.0281±.0177	1.000	0.0332±.0165	1.000	0.0491±.0084	1.000
s3_e2	0.0289±.0182	1.000	0.0321±.0164	1.000	0.0525±.0136	1.000
s4_e1	0.0332±.0233	1.000	0.0336±.0223	1.000	0.0420±.0111	1.000
s4_e2	0.0332±.0225	1.000	0.0345±.0222	1.000	0.0452±.0093	1.000

Table A2: Cross-parent transfer probe (unchanged from the per-parent study): the July donor controller evaluated on each recipient parent. Final Hamming and survival on held-out damage seeds; transfer is a penalty in 5/5 recipients, lethal in 2.

Recipient seed	Final Hamming	Survival
0	0.036	1.00
1	0.249	0.00 (lethal)
2	0.434	0.00 (lethal)
3	0.100	0.45
4	0.041	1.00

Table A3: Full 5×5 cross-parent transfer matrix among the five defense parents, both controller replicas (top: e1; bottom: e2). Rows: donor controller; columns: recipient parent. Cells give mean final Hamming / worst-case survival over 3 condition seeds × 8 held-out damage seeds (24 rollouts per cell). Diagonal (bold): recipient's own controller (survival 1.00 throughout). †lethal (survival 0.00). Across the 40 off-diagonal cells: 23 harmful, 15 indistinguishable from zero-output, 2 beneficial; the beneficial cells are exclusively s4→s1, the tonic-aligned pair (cosine +0.99), in both replicas.

e1 donor	recip. s0	recip. s1	recip. s2	recip. s3	recip. s4
s0	**0.0315**	0.470/0.00†	0.097/0.38	0.028/1.00	0.052/0.88
s1	0.035/1.00	**0.0306**	0.023/1.00	0.049/1.00	0.035/1.00
s2	0.029/1.00	0.043/0.88	**0.0201**	0.033/1.00	0.092/0.50
s3	0.035/1.00	0.036/0.88	0.208/0.00†	**0.0295**	0.044/1.00
s4	0.034/1.00	0.030/1.00	0.023/1.00	0.044/1.00	**0.0340**
e2 donor					
s0	**0.0291**	0.108/0.25	0.056/0.88	0.032/1.00	0.039/1.00
s1	0.032/1.00	**0.0292**	0.025/1.00	0.049/1.00	0.032/1.00
s2	0.041/1.00	0.074/0.62	**0.0219**	0.033/1.00	0.113/0.25
s3	0.028/1.00	0.058/0.75	0.239/0.00†	**0.0262**	0.028/1.00
s4	0.030/1.00	0.030/1.00	0.020/1.00	0.033/1.00	**0.0336**

Tonic transplant. Replacing each donor controller by its realized mean m_t injected as a constant (same protocol, both replicas, 3 condition seeds) reproduces the controller's outcome in 36/40 off-diagonal cells and 10/10 diagonal cells (46/50 overall; Figure 1B), including every lethal cell, where the transplant kills exactly as the controller does. The four mismatches are magnitude differences inside already-lethal cells. Under the tested regime, the controllers' small dynamic residual did not measurably contribute, on-parent or off (full CSVs in the repository).

AUC mirrors the same ordering in every seed (full CSVs in the repository).

B Controller-output (m_t) diagnostics

Table A4: Tonic output of the evolved controllers: per-channel mean of m_t, averaged over the two independent evolutions per seed. All ten controllers sit at nonzero, parent-distinct offsets; within-rollout per-channel std of m_t is 0.0002–0.0046 everywhere: flat, tonic output with no lesion-locked response in 10/10 runs.

Seed	mean(m) per channel (avg. of 2 runs)		
0	-0.021	$+0.002$	$+0.027$
1	$+0.030$	$+0.012$	-0.026
2	-0.026	-0.032	-0.015
3	$+0.001$	$+0.032$	-0.010
4	$+0.028$	$+0.007$	-0.025

The tonic vector alone does not predict benefit: seeds 1 and 4 emit nearly identical tonic vectors (cosine similarity 0.993, Euclidean distance 0.0065), yet only seed 1 benefits from its controller. The

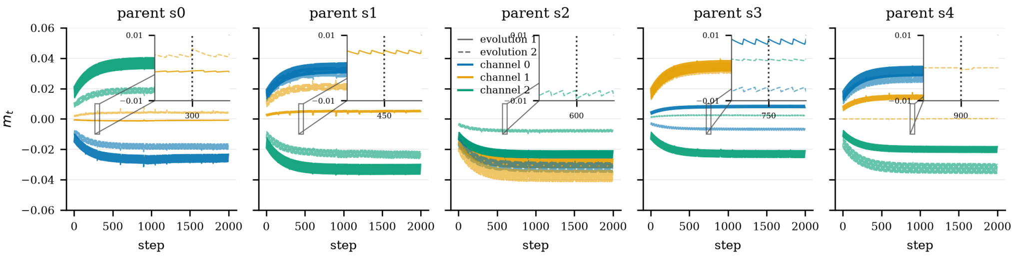

Figure A1: Controller output m_t over a full evaluation rollout, per parent seed, both independent evolutions overlaid (solid/dashed), one line per modulator channel. After a brief initial transient while the perceptual state settles, every controller holds a constant, parent-distinct offset through all lesion events: tonic calibration, not an event-locked policy. Note that the two independent evolutions of the same parent sometimes settle at slightly different setpoints.

benefit arises from the interaction between a tonic calibration and parent-specific local dynamics, not from a distinctive chemical operating point.

Tonic alignment structures transfer. All lethal instances pair strongly negative tonic-vector cosines (s3→s2: -0.61; s0→s1: -0.93), while the only beneficial transfers, beating zero-output and matching the own controller, occur exclusively within the tonic-aligned pair s4→s1 (cosine $+0.99$, both replicas; adjacent to the diagonal in Figure 1). Across all 40 cells, tonic cosine correlates with transfer penalty ($r = -0.30$) and survival ($r = +0.34$); alignment is not sufficient, as some benign cells carry cosines to -0.95.

C Pilot: single-parent evaluation hides parent-locking

A three-parent pilot (2026-08-16) first exposed the confound. The single-parent controller from our original five-condition study transferred lethally to two of three sibling parents (survival 0.00, final Hamming 0.249 and 0.434), while zero-output channel parents beat the $K{=}0$ baseline in $3/3$ seeds ($0.029/0.034/0.020$ vs $0.034/0.177/0.028$). The original table's headline gap was therefore part parent-training effect and part parent-seed luck, motivating this study.

D Model and training details

The grid state $x_t \in [0, 1]^{96\times96\times16}$ holds 16 channels; the last four are RGBA with alpha last. Perception uses three fixed kernels (identity, Sobel-x, Sobel-y) per channel, giving $p_t \in \mathbb{R}^{48}$:

$$p * t = \big(K * \mathrm{id}_x * t,\ K * \mathrm{S\_}x_x * t,\ K * \mathrm{S\_}yx_t\big). \tag{1}$$

The update MLP maps $(48{+}K)$ inputs to 128 hidden units (ReLU) and back to 16 channel increments, with the final layer zero-initialized. Cells update with probability 0.5; a cell is alive when its alpha exceeds 0.1 within its $3{\times}3$ max-pooled neighborhood; dead cells are masked to zero.

Tonic and phasic modulator dynamics:

$$m * t^{(\mathrm{tonic})} = \alpha\, m * t - 1^{(\mathrm{tonic})} + (1-\alpha)\, c * t, \qquad m_t^{(\mathrm{phasic})} = m * t - 1^{(\mathrm{phasic})} \cdot e^{-\Delta t/\tau}, \tag{2}$$

with $\alpha = 0.95$, $\tau = 20$, and the injected level the clipped sum, broadcast to all cells and concatenated to perception (input dim $48 + 3 = 51$).

Parents train on a $40{\times}40$ emoji target zero-padded to the canvas, with pool-based training (pool 1024, batch 8, worst-in-batch reseeding), random square-erasure damage mid-episode, MSE loss on RGBA, and Adam at 10^{-3} for 8000 steps. A first attempt at $2{\times}10^{-3}$ diverged to NaN near step 5532 via a late-stage loss-spike cascade; 10^{-3} absorbs the identical spike.

E Perception-radius sweep

A single-lesion sweep (radii $\{2, 4, 8, 16\}$, single- and multi-site, with and without modulator channels) maps the boundary of local repair. Recovery is near-perfect for small lesions and degrades once lesions exceed the effective perception radius. Two failure signatures motivate this work: severed fragments that survive but cannot reattach, die, or decide what to grow into, and persistent half-alpha scar tissue at wound sites. Both are information failures: the misbehaving cells are exactly the cells cut off from global context.

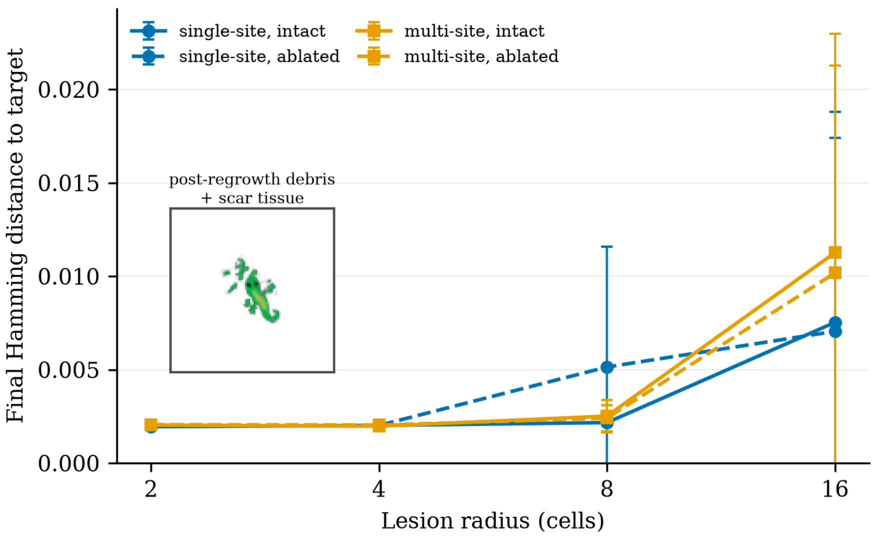

Figure A2: E1 lesion sweep. Final Hamming distance as a function of lesion radius and kind, with and without modulator channels (mean $\pm$ SD over 5 damage seeds). Inset: post-regrowth debris and scar tissue.

F Calibration probes

Collapse probe. The target mask is only 4.6% alive, so a fully-dead grid scores Hamming ≈ 0.046, close to a struggling organism's ≈ 0.02. We verified empirically that the all-dead state scores 0.0461 against struggling-neutral's 0.0205: death loses by $2.2\times$, so the alive-fraction floor is disabled (0.0). Notably, the runbook-default floor of 0.3 would have *penalized every healthy candidate* (a correct lizard lives at ≈ 0.057 alive), pushing evolution toward overgrowth, a case where calibrating from the experiment's own data overrode a plausible default.

Initial step-size probe. Perturbing the neutral controller by Gaussian noise at increasing σ (25 samples each, hard-regime evaluation) showed: $\sigma \leq 0.01$ stays healthy (some samples beat neutral); $\sigma = 0.05$ is bimodal; $\sigma = 0.3$ lands *every* sample in saturated overgrowth (alive $\rightarrow 0.96$). A first evolution run with the library-default $\sigma_0 = 0.3$ froze at fitness 0.150 from generation 2 onward, $7\times$ worse than doing nothing, because CMA-ES was ranking overgrowth-degree noise. $\sigma_0 = 0.01$ brackets the neutral point and the same budget converged smoothly (Figure A3).

G Evolution objective

We evolve the controller with CMA-ES [12] (Evosax implementation [13]), population 64, initial step size σ_0=0.01, 300 generations, neutral (all-zero) controller as the initial mean. Fitness is the event-weighted mean Hamming distance to the target alpha mask over full rollouts on the eight train damage seeds:

$$f(\theta) = \frac{1}{\sum *t w_t} \sum *t = 1^T w * t\, H_t(\theta), \qquad w_t = \exp\Big(-\frac{t - \tau * \mathrm{last}(t)}{\tau * w}\Big), \tag{3}$$

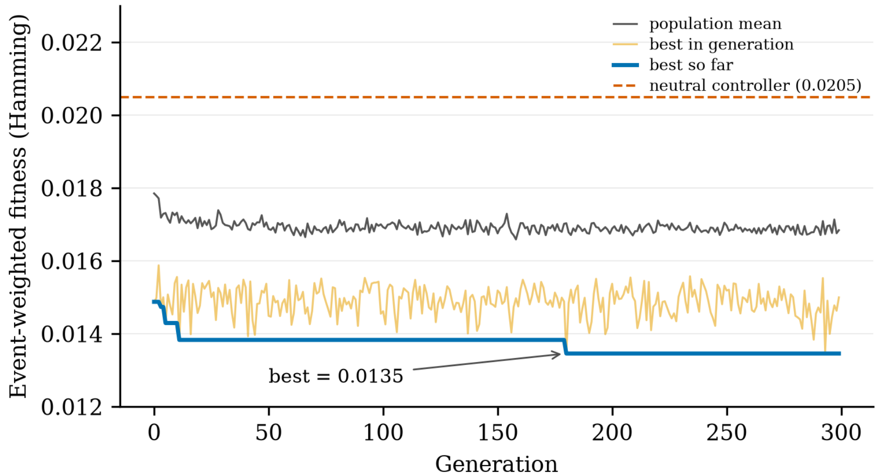

Figure A3: CMA-ES fitness over 300 generations (σ_0=0.01). Dashed line: neutral controller (0.0205). Best evolved fitness: 0.0135, a 34% improvement with no stall.

where $\tau * \mathrm{last}(t)$ is the most recent lesion step and $\tau_w = 150/3 = 50$. The recency kernel resets at each lesion, so slow repair is expensive even when endpoint Hamming is identical, so it de-saturates the metric and prices repair speed directly. Fitness is not the reported metric: all results report unweighted trajectory statistics on held-out seeds. Two calibration controls make this landscape interpretable: a collapse probe (verifying the objective cannot be gamed by alpha suppression) and an initial step-size ablation (showing σ_0=0.3 saturates the first population in overgrowth where no repair signal exists), both detailed in Appendix F.

H Single-parent five-condition study

Five conditions share the same protocol, horizon, and damage seeds: **closed-loop** (evolved controller); **static** (the evolved controller evaluated once at t=0 and held constant, isolating temporal modulation from the evolved tonic level itself); **constant** (fixed tonic, grid-searched over $\{-1, -0.5, 0, 0.5, 1\}$ on train seeds); **random** (uniform $[-1, 1]$ each step, actuation-matched noise); and **no modulation** (K=0 parent). All experiments ran on a single rented A100; parent training took ≈20 min/model, evolution ≈2.5 h, total project cost under $15.

Modulation helps substantially (RQ1). Relative to no modulation, modulated conditions recover 2.2× more completely (final Hamming 0.028–0.030 vs. 0.063) and sustain $\approx 4\times$ less cumulative damage (AUC 0.016–0.017 vs. 0.064). The wounds exceed the perception radius, so this improvement is attributable to the broadcast channel, the only non-local pathway, which repairs damage that local rules cannot diagnose.

A regime boundary, not a failed ablation (RQ2). The three modulated conditions are statistically indistinguishable: final Hamming spans 0.028–0.030, half-life 6.4–7.2 steps, all differences within noise (pairwise effect sizes < 0.15). We read this as a measurement of the *regime*, not a failure of the controller: the damage process is stationary and memoryless (events are i.i.d. in size, number, and position), so the optimal release policy is time-invariant and a fixed tonic gain is structurally adequate. There is no temporal structure for adaptive scheduling to exploit. Evolution's measured contribution is automatic discovery of the near-optimal tonic level in 2.5 hours from a neutral start, without the hand search the constant condition required.

Random modulation is lethal. Random actuation kills every run (survival 0.00, final Hamming 0.825): pilot probes show constant levels alone swing mean Hamming from 0.003 to 0.93. The modulator channels are a high-gain control pathway, not a free architectural bonus; the identity of the release schedule determines whether the organism lives.

Metric artifacts of long-horizon evaluation (RQ3). Two artifacts surfaced only under repeated damage. First, repair half-life is defined per lesion relative to each post-lesion jump; on the baseline's chronically damaged morphology, new lesions often land on already-degraded regions, producing no measurable jump and scoring as zero half-life, so the baseline's 2.7 steps does *not* indicate fast repair. Second, residual Hamming in all conditions is partly attributable to locomotion drift during regeneration rather than permanent damage; the metric conflates spatial translation with morphological error. Both are documented so that future benchmark users can avoid them.

I Reproducibility

Parents train in $\approx$20–30 min each on one A100; the defense study ran two independent 300-generation controller evolutions per parent (10 total), each $\approx$2.5–3 h on the study A100, for a total defense-study cost of $\approx$\$25 of rented GPU time. The transfer matrix is evaluation-only over existing artifacts: 40 off-diagonal evaluations, $\approx$2 minutes of A100 time per full 5×5 replica matrix. Preregistration, analysis script, per-seed artifacts (controllers, parents, trajectories, m_t series), and the mechanical outcome output are in the repository, timestamped before the results. Code will be released upon publication. Code mirror: `https://anonymous.4open.science/r/neuromodulation-8C57/`.